# Maintenance Schedule

Diesel Engine
**10V1600A60**
**12V1600A60**
**Application Group 5B**

**MS50164/01E**

Power. Passion. Partnership.

| Engine model | kW/cyl. | Application group |
|---|---|---|
| 10V1600A60 | 57 kW/cyl. | 5B, Continuous operation, variable |
| 12V1600A60 | 57 kW/cyl. | 5B, Continuous operation, variable |

*Table 1: Applicability*

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# 1 Maintenance Schedule

## 1.1 Preface

In the case of emission-certified products, nonobservance of the maintenance specifications can result in a violation of the statutory emission regulations. Modification or removal of mechanical or electronic components or the installation of additional components as well as the execution of calibration processes that might affect the emission characteristics of the product are prohibited by emission regulations. Components relevant to emissions, e.g. control units, sensors, cylinder heads, injectors and exhaust flaps can only be serviced, replaced or repaired if components approved by MTU, or equivalent components, are used.

The MTU maintenance system is based on a preventive maintenance concept. Preventive maintenance facilitates advance planning and ensures a high degree of equipment availability. The maintenance intervals and required maintenance scopes are based on operational experience and must therefore be regarded as recommendations. In the event of difficult operational and ambient conditions, additional maintenance tasks and/or changes to the maintenance intervals may be necessary.

The intervals at which the maintenance tasks have to be carried out are specified in operating hours and time periods. The limit which occurs first is applicable.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational checks and maintenance work that does not require disassembly of the product.
QL2: Component replacement (corrective, and thus not an element of the maintenance schedule).
QL3: Maintenance work that requires partial disassembly of the product.
QL4: Maintenance work that requires complete disassembly of the product.

### Supplementary information on maintenance and preservation:

The intervals at which fluids and lubricants should be changed and details regarding corrosion-proofing can be obtained from the relevant specifications of the component manufacturers and the MTU Fluids and Lubricants Specifications. The latest version for drive systems is available online at: http://www.mtu-online.com, and for power generation systems at: http://www.mtuonsiteenergy.com.

With regard to the maintenance of all components not listed in this maintenance schedule, the manufacturer's specifications apply.

This maintenance schedule does not apply to engines with an EPA/CARB emission certificate.

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## 1.2 Allocation matrix application group 5B

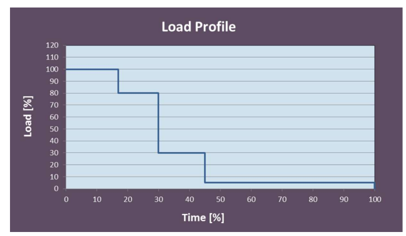
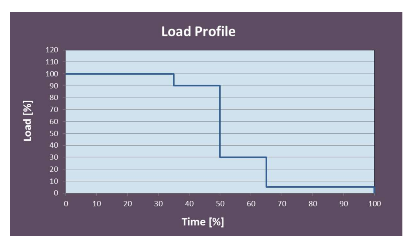
| Load profile | | Load factor % | Load indicator % | 10/12V1600C60/A60 | | | | |
|---|---|---|---|---|---|---|---|---|
| Load % | Time % | | | 0 - 60 kW/cyl. | \| | \| | \| | |
| 110 (100-110) | – | 0 - 31 | ≤ 17 | 21,000 | – | – | – | Standard   |
| 100 (90-100) | 12 | | | | | | | |
| 90 (80-90) | 12 | | | | | | | |
| 80 (65-80) | – | | | | | | | |
| 65 (50-65) | – | | | | | | | |
| 50 (30-50) | – | | | | | | | |
| 30 (5-30) | 16 | | | | | | | |
| Idle (0-5) | 60 | | | | | | | |
| 110 (100-110) | – | 32 - 35 | ≤ 19 | 18,000 | – | – | – |   |
| 100 (90-100) | 17 | | | | | | | |
| 90 (80-90) | – | | | | | | | |
| 80 (65-80) | 13 | | | | | | | |
| 65 (50-65) | – | | | | | | | |
| 50 (30-50) | – | | | | | | | |
| 30 (5-30) | 15 | | | | | | | |
| Idle (0-5) | 55 | | | | | | | |
| 110 (100-110) | – | 36 - 45 | ≤ 32 | 15,000 | – | – | – |   |
| 100 (90-100) | 26 | | | | | | | |
| 90 (80-90) | 15 | | | | | | | |
| 80 (65-80) | – | | | | | | | |
| 65 (50-65) | – | | | | | | | |
| 50 (30-50) | – | | | | | | | |
| 30 (5-30) | 12 | | | | | | | |
| Idle (0-5) | 47 | | | | | | | |
| 110 (100-110) | – | 46 - 55 | ≤ 41 | 15,000 | – | – | – |   |
| 100 (90-100) | 35 | | | | | | | |
| 90 (80-90) | 15 | | | | | | | |
| 80 (65-80) | – | | | | | | | |
| 65 (50-65) | – | | | | | | | |
| 50 (30-50) | – | | | | | | | |
| 30 (5-30) | 15 | | | | | | | |
| Idle (0-5) | 35 | | | | | | | |

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*Table 2: Allocation matrix application group 5B*

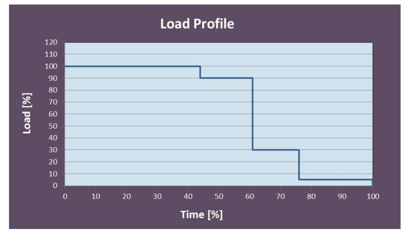
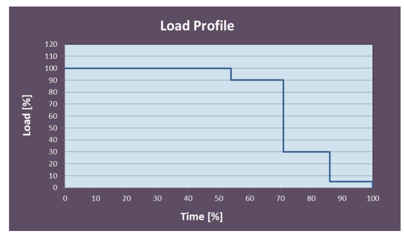
| 10/12V1600C60/A60 | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Load profile | | Load factor % | Load indicator % | 0 - 60 kW/cyl. | – | – | – | |
| Load % | Time % | | | | | | | |
| 110 (100–110) | – | 56 - 65 | ≤ 51 | 12,000 | – | – | – |  |
| 100 (90–100) | 44 | | | | | | | |
| 90 (80–90) | 17 | | | | | | | |
| 80 (65–80) | – | | | | | | | |
| 65 (50–65) | – | | | | | | | |
| 50 (30–50) | – | | | | | | | |
| 30 (5–30) | 15 | | | | | | | |
| Idle (0–5) | 24 | | | | | | | |
| 110 (100–110) | – | 66 - 75 | ≤ 61 | 9,000 | – | – | – |  |
| 100 (90–100) | 54 | | | | | | | |
| 90 (80–90) | 17 | | | | | | | |
| 80 (65–80) | – | | | | | | | |
| 65 (50–65) | – | | | | | | | |
| 50 (30–50) | – | | | | | | | |
| 30 (5–30) | 15 | | | | | | | |
| Idle (0–5) | 14 | | | | | | | |

*Table 3: Allocation matrix application group 5B*

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# 1.3 Maintenance schedule matrix 21,000 h

## 0-21,000 Operating hours

| Task | Limit | Operating hours [h] | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Limit | Daily | 1,000 | 2,000 | 3,000 | 4,000 | 5,000 | 5,250 | 6,000 | 7,000 | 8,000 | 9,000 | 10,000 | 10,500 | 11,000 | 12,000 | 13,000 | 14,000 | 15,000 | 15,750 | 16,000 | 17,000 | 18,000 | 19,000 | 20,000 | 21,000 | | | | | |
| Engine | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| W1008 | 2 a | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| W0500 | - | X | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| W0501 | - | X | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| W0506 | - | X | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| W0507 | - | X | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| W1001 | 2 a | | X | X | X | X | X | | X | X | X | X | X | | X | X | X | X | X | | X | X | X | X | X | X | | | | | |
| W1675 | 2 a | | X | X | X | X | X | | X | X | X | X | X | | X | X | X | X | X | | X | X | X | X | X | X | | | | | |
| W1019 | - | | | X | | X | | | X | | X | | X | | | X | | X | | | X | | X | | X | | | | | | |
| W1716 | - | | | | X | | | | X | | | X | | | | X | | | X | | | | X | | | X | | | | | |
| W1207 | 2 a | | X | | X | | | | X | | | X | | | | X | | | X | | | | X | | | X | | | | | |
| W1003 | 2 a | | | | X | | | | X | | | X | | | | X | | | X | | | | X | | | X | | | | | |
| W8638 | - | | | | | | | X | | | | | | X | | | | | | X | | | | | | X | | | | | |
| W1525 | 6 a | | | | | | | X | | | | | | X | | | | | | X | | | | | | X | | | | | |
| W1526 | 6 a | | | | | | | X | | | | | | X | | | | | | X | | | | | | X | | | | | |
| W1636 | 9 a | | | | | | | | | X | | | | | | | | X | | | | | | | | X | | | | | |
| W1006 | 9 a | | | | | | | | | X | | | | | | | | X | | | | | | | | X | | | | | |
| W1298 | 9 a | | | | | | | | | X | | | | | | | | X | | | | | | | | X | | | | | |
| W1178 | 9 a | | | | | | | | | X | | | | | | | | X | | | | | | | | X | | | | | |
| W1660 | - | | | | | | | | | | | | | X | | | | | | | | | | | | X | | | | | |
| W1812 | 5 a | | | | | | | | | | | | | X | | | | | | | | | | | | X | | | | | |
| W1055 | 6 a | | | | | | | | | | | | | X | | | | | | | | | | | | X | | | | | |
| W8637 | 6 a | | | | | | | | | | | | | X | | | | | | | | | | | | X | | | | | |
| W1041 | 9 a | | | | | | | | | | | | | X | | | | | | | | | | | | X | | | | | |
| W1672 | 9 a | | | | | | | | | | | | | | | | | | | | | | | | | X | | | | | |
| W3148 | 18 a | | | | | | | | | | | | | | | | | | | | | | | | | X | | | | | |

w = weeks
m = months
a = years

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